

FAB-500

The **FAB-500** is a Soviet-designed 500-kilogram (1,100 lb) general purpose air-dropped bomb with a high-explosive warhead, primarily used by the Russian Aerospace Forces, former Soviet republics and customer countries. The original M-54 model was rolled out in 1954, shaped for internal carriage by heavy bombers, a low-drag M-62 version in 1962 was intended for fighter bomber external hardpoint carriage.^{[2][3]} The bomb is unguided, features a single nose fuze, and is compatible with most models of Soviet aircraft.^[4]

Operational history

The FAB-500 was widely employed over Afghanistan by Soviet and allied Afghan forces during the 1980s and saw use during the Syrian civil war where it was carried by both Russian and Syrian warplanes.^{[5][6][7]} The M62 variant of the FAB-500 was used by Russian military forces in the 2022 Russian invasion of Ukraine.^[8] On 13 March 2022 and 14 May 2022, FAB-500 bombs were found in Ukrainian cities Chernihiv^[9] and Odesa.^[10]

In March 2023, Russian Su-35s have launched a number of FAB-500M-62, wreckage of which indicates that these have been fitted with a glide kit, involving pop-out wings. It's unknown whether it has an internal navigation system, or has been fitted with wings simply to extend its range to up to 70 km.^[11] It also is believed to give Russian aircraft a stand-off ability to hit Ukrainian targets without risking exposure to Ukrainian air defences.^{[12][13][14]}

As of May 2023, converted FAB-500 glide bombs continued to be used by Russia in Ukraine, with up to 20 of them being dropped every day, and Ukrainian air defences lack the capability to intercept them.^{[11][15]}

Operational envelope (FAB-500 M-62)

- Release altitude: 570–12,000 metres (1,870–39,400 ft)

FAB-500 M-62 General Purpose (GP) Bomb ^[1]	
	
FAB-500 M-62, first in foreground	
Type	High-drag and low-drag general-purpose bomb
Place of origin	<u>Soviet Union</u>
Production history	
Produced	1954–present (M-54); 1962–present (M-62)
Variants	<u>KAB-500</u>
Specifications	
Mass	500 kilograms (1,100 lb)
Length	2,470 millimetres (97.2 in)
Diameter	400 millimetres (15.7 in)
Filling	High explosive
Filling weight	300 kilograms (660 lb)



FAB-500 M-54 General-purpose bomb (high drag version)

- Release speed: 500–1,900 kilometres per hour (311–1,180 mph)^[16]

Variants

- **FAB-500 M-54** (ФАБ-500 М-54) – 1954 model, original high-drag model intended for internal carriage on heavy bombers, featuring a ballistic ring on the nose of the bomb to act as a vortex generator to aid the bomb's stabilizers.^{[17][18]}
- **FAB-500 M-62** (ФАБ-500 М-62) – 1962 model, low-drag model designed for external carriage on hardpoints on fighter-bombers.^{[19][20]}
- **FAB-500 M-62T** – heat-resistant, capable of withstanding high-temperature aerodynamic heating, arising during flight at high speeds. Developed for the MiG-25RB.^[20]
- **FAB-500 M-62 MPK** (ФАБ-500 М-62 с МПК)
- **FAB-500T** (ФАБ-500Т)^[21]
- **FAB-500TA** (ФАБ-500ТА)
- **FAB-500ShN** (ФАБ-500ШН) – parachute retarded thin cased bomb^[19]
- **FAB-500ShL** (ФАБ-500ШЛ) – parachute retarded high explosive bomb^{[19][22]}
- **FAB-500 M46** (ФАБ-500 М46)
- **FAB-500 M44** (ФАБ-500 М44)
- **FAB-500 M43** (ФАБ-500 М43)
- **FAB-500-300** (ФАБ-500-300) – thick cased high explosive
- **FAB-500TS** (ФАБ-500ТС) – thick cased high explosive and fragmentation^{[21][22][23]}
- **FAB-500TSM** (ФАБ-500ТСМ)
- **OFAB-500** (ОФАБ-500) – general purpose high explosive fragmentation bomb (contains 3 x 150 kg submunitions)^[24]
- **OFAB-500U** (ОФАБ-500У) – delayed action fragmentation bomb
- **OFAB-500ShR** (ОФАБ-500ШР) – parachute retarded fragmentation bomb



up to 18 FAB-500 M-54 general-purpose bombs on two underwing pylons and internal bay of a Tu-22M



FAB-500 with gliding system (Prototype)

See also

- FAB-250
- Mark 83 bomb – American counterpart

References

1. "FAB-500 M-62 high-explosive bomb" (<http://roe.ru/eng/catalog/aerospace-systems/air-bomb/s/fab-500-m-62/>). *Rosoboronexport*.
2. U.S.S.R. BOMB, FAB-250 M46, General purpose (https://www.jmu.edu/cisr/_pages/research/afghanistan-oig/01-bomb.pdf)
3. "Russian Aviation Bombs" (<https://www.globalsecurity.org/military/world/russia/bombs.htm>).

4. "FAB-500" (http://www.deagel.com/Defensive-Weapons/FAB-500_a003491001.aspx). *Guide to Military Equipment and Civil Aviation*. 15 July 2017. Retrieved 15 March 2018.
5. "Afghanistan Ordnance Identification Guide" (<https://core.ac.uk/download/pdf/153206738.pdf>) (PDF). Center for International Stabilization and Recovery. 2004.
6. "DFID funded EOD team defuses 500kg aircraft bomb in Herat city" (<https://www.halotrust.org/latest/halo-updates/news/dfid-funded-eod-team-defuses-500kg-aircraft-bomb-in-herat-city/>). *The HALO Trust*.
7. "The Su-22 Is Syria's War-Weary Warhorse" (<https://warisboring.com/the-su-22-is-syrias-war-weary-warhorse/>). *War Is Boring*. 15 February 2017. Retrieved 15 March 2018.
8. "Three FAB-500 aerial bombs found in Chernihiv" (<https://news24online.com/news/world/russia-ukraine-war-three-fab-500-aerial-bombs-found-chernihiv-219eab39/>). *news 24*. 15 February 2017. Retrieved 5 March 2022.
9. "In Chernihiv, three unexploded 500kg bombs discovered in residential areas" (<https://www.ukrinform.net/rubric-ato/3428374-in-chernihiv-three-unexploded-500kg-bombs-discovered-in-residential-areas.html>). *ukrinform.net*. 13 March 2022. Retrieved 10 September 2022.
10. "Rescuers defuse two 500-kg aircraft bombs in Odesa Oblast" (<https://english.nv.ua/nation/rescuers-defuse-two-500-kg-aircraft-bombs-in-odesa-oblast-ukraine-news-50242185.html>). *english.nv.ua*. 14 May 2022. Retrieved 10 September 2022.
11. "Russia deploys up to 20 guided bombs each day – Ukraine's Air Force" (<https://news.yahoo.com/russia-deploys-20-guided-bombs-080226488.html>). *Yahoo*. 4 April 2023. Retrieved 7 April 2023.
12. "Mystery Surrounds Appearance of Russian Aerial Bombs Fitted with Wings" (<https://www.kyivpost.com/post/15165>). *kyivpost*. 31 March 2023. Retrieved 5 April 2023.
13. "UMPK guided bomb" (<https://www.globalsecurity.org/military/world/russia/umpk.htm>). *globalsecurity*. 27 March 2023. Retrieved 5 April 2023.
14. "Russia uses its own kind of "smart" bombs in Sumy Oblast" (<https://news.yahoo.com/russia-uses-own-kind-smart-135232155.html>). *Yahoo*. 24 March 2023. Retrieved 5 April 2023.
15. Gettleman, Jeffrey; Schmitt, Eric (26 May 2023). "Russia's Old Bombs Elude Ukraine's Modern Defenses" (<https://news.yahoo.com/russia-old-bombs-elude-ukraine-174935901.html>). *Yahoo*. Retrieved 27 May 2023.
16. "High-explosive bomb FAB-500 M-62 | Catalog Rosoboronexport" (<http://roe.ru/eng/catalog/aerospace-systems/air-bombs/fab-500-m-62/>).
17. Gordon, Yefim (2004). *Soviet / Russian aircraft weapons : since World War Two*. Hinkley: Midland Publishing. p. 158. ISBN 1857801881. OCLC 56650196 (<https://www.worldcat.org/oclc/56650196>).
18. "ФАБ-500М-54" (<http://aviaros.narod.ru/fab50054.htm>). *aviaros.narod.ru*. Retrieved 8 March 2022.
19. "United Nations Mine Action Service (UNMAS) – Ammunition Identification Guide – Libya" (<https://bulletpicker.com/pdf/Ammunition-Identification-Guide-Lybia.pdf>) (PDF). 2015.
20. "ФАБ-500М-62" (<http://aviaros.narod.ru/fab50062.htm>). *aviaros.narod.ru*. Retrieved 8 March 2022.
21. "FAB-250 General purpose Bomb" (<https://www.globalsecurity.org/military/world/russia/fab-250.htm>). *www.globalsecurity.org*. Retrieved 2022-03-07.
22. "Авиационное бомбовое вооружение" (http://www.militaryparitet.com/nomen/russia/aviabomb/data/ic_nomenrussiaaviabomb/16/). *www.militaryparitet.com*. Retrieved 7 March 2022.
23. "ФАБ-250ТС/500ТС" (<http://www.airwar.ru/weapon/ab/fab-250ts.html>). *www.airwar.ru*. Retrieved 7 March 2022.

24. "General-purpose HE fragmentation bomb OFAB-500U | Catalog Rosoboronexport" (<http://roe.ru/eng/catalog/aerospace-systems/air-bombs/ofab-500u/>). *roe.ru*. Retrieved 13 March 2022.
-

Retrieved from "<https://en.wikipedia.org/w/index.php?title=FAB-500&oldid=1177823494>"

■